

Regular Article

Risk of Insomnia Attributable to β -Blockers in Elderly Patients with Newly Diagnosed Hypertension

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Summary: Use of β -blockers may cause insomnia and central nervous system and/or psychological side effects, but data are limited on the relative risks of insomnia among β -blockers. This retrospective cohort study used Taiwan's National Health Insurance claims database from 2003 to 2007, where 4,063 patients aged above 65 years with newly diagnosed hypertension and treated with β -blockers were followed for 1 year. The primary endpoint was a new insomnia event within 30 days of treatment initiation. Adjusted odds ratios of insomnia were obtained by logistic regressions, controlling for baseline risk factors of insomnia. Using propranolol therapy as the reference, the adjusted odds ratio (95% confidence interval) for the insomnia risk was 0.47 (0.35–0.63) for non-propranolol users, 0.31 (0.19–0.50) for bisoprolol, and 0.46 (0.33–0.66) for atenolol. Compared to the patients using non-selective β -blockers, the adjusted odds ratio was 0.48 (0.36–0.64) for those using selective β_1 -blockers. Additionally, the adjusted odds ratio was 0.72 (0.53–0.96) for β -blockers with low lipophilicity when compared to those with high lipophilicity. The use of bisoprolol and atenolol was associated with the lowest risk of insomnia in elderly patients, as compared to propranolol. β -Blockers with high selectivity in β_1 -receptors and/or low lipophilicity were associated with a lower risk of insomnia.

Keywords: β -blocker; insomnia; elderly; hypertension; pharmacoepidemiology

Introduction

β -Blockers are recommended as first-line agents to treat hypertension with compelling indications, such as post-myocardial infarction and coronary disease.¹⁾ Because of the indications, many β -blockers are used by the elderly population. As such, although these medications have an essential role in the clinical management of hypertension and cardiovascular diseases, previous clinical studies indicated that central nervous system (CNS) and/or psychological side effects (including insomnia) of the drugs might have impeded their use, especially among elderly patients, who are more vulnerable to CNS side effects.^{2–5)} Although limited information is available on possible association between β -blockers

and CNS side effects, pindolol and propranolol were considered to cause more CNS side effects in the clinical trials due to higher brain concentration, as observed in a pharmacokinetic study.⁶⁾ However, though insomnia is commonly listed in the package inserts for β -blockers, to our knowledge, there has been no study reporting either on the incidence of β -blocker-induced insomnia among the elderly population in real-world practice settings, or on the relative risks of this side effect across the entire spectrum of β -blockers.

The published literature indicates that insomnia is often associated with significant medical and psychiatric morbidity in the elderly population, not to mention the adverse impacts on patients' quality of life.^{7–9)} The consequences of insomnia from longitudinal studies include decreased physical function,¹⁰⁾ falls/

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fracture,¹¹⁾ cardiovascular disease and mortality.^{12,13)} One meta-analysis showed that patients with hypnotic-treated insomnia were associated with increased risk of cognitive impairment and psychomotor-type adverse events (including dizziness, loss of balance or falls) than with a placebo.¹⁴⁾

Previous clinical trials on some β -blockers have reported insomnia events when profiling the side effects of the treatments.^{15,16)} No significant difference in the overall incidence of insomnia events was found between the β -blockers, probably due to the limited sample size of these studies.¹⁶⁾ In a randomized, double-blind, crossover study on hypertensive patients receiving metoprolol, atenolol or a placebo, metoprolol was associated with a significant increase in the incidence of sleep disturbance.¹⁷⁾

One community-based study showed that 42% of adults aged 65 years and older had difficulty falling asleep and staying asleep in the USA.¹⁸⁾ Studies in EU countries found the prevalence of insomnia to be between 19% and 37% in the general population, with higher prevalence among the elderly.^{19–21)} In Taiwan, the prevalence of insomnia among the elderly population was 11.14%,²²⁾ and that of hypertension was 58.1%.²³⁾ Since β -blockers were one of the most frequently prescribed antihypertensive agents in Taiwan, second only to calcium channel blockers,^{24,25)} it is possible that some elderly patients who received β -blockers for hypertension control suffered from the insomnia side effect of the treatment. Using the national population databases, the objective of this study was to determine the relative risks of insomnia among β -blockers among elderly patients with newly diagnosed hypertension.

Methods

Data source: This retrospective cohort study analyzed the claims databases of National Health Insurance (NHI) in Taiwan from January 1, 2003 through December 31, 2007. The databases provide information on ambulatory service, hospitalization, and medication use for all enrollees of the NHI. The NHI is a single-payer program that offers universal coverage of health care, including comprehensive medical services and prescription drugs, to all citizens in Taiwan. The NHI program enrolls more than 99% of the Taiwanese population and contracts with 92.47% of all hospitals and clinics²⁶⁾ in Taiwan. The recorded information in the NHI database includes the patient's characteristics (*e.g.*, age, sex), ICD-9-CM codes of diagnoses and procedures, and prescriptions with dispensing date, drug codes, dose, quantity, and number of days of supply. Patient identifiers have been encrypted for privacy protection, but all data sets could be linked together with the unique and anonymous identifiers created by data providers for research purpose.

Study population: Patients 65 years or older with newly diagnosed hypertension between January 1, 2004 and December 31, 2006 were included in this study. The index date was defined as the date of first newly diagnosed hypertension. To be eligible for the study, patients need to be prescribed with β -blockers upon the diagnosis of hypertension; in addition, information on other antihypertensive drugs was also extracted, including diuretics, calcium channel blockers, angiotensin converting enzyme inhibitors/angiotensin II receptor blockers (ACEI/ARB), α -blockers, and vasodilators acting on smooth muscle. Hypertension was identified by using ICD-9-CM codes of 401.xx, 402.xx, 403.xx, and 404.xx. The baseline period was defined as one year prior to the index date, and the follow-up period was one year from the index date.

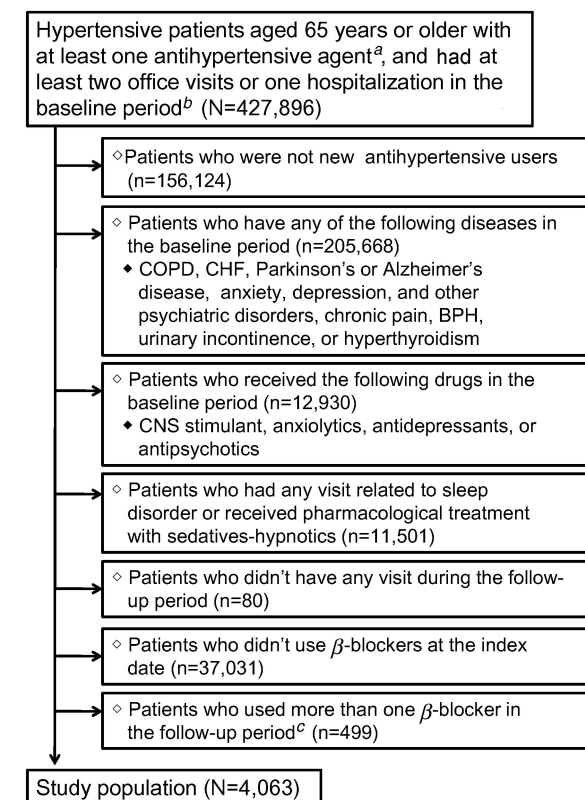
Patients were excluded if they had a previous hypertension diagnosis and/or used above mentioned antihypertensive agents except acting on the angiotensin system before the index date. Prior use of ACEI/ARB was not excluded because these medications might be indicated for renal protection²⁷⁾ and/or cardioprotection²⁸⁾ and might not necessarily be related to antihypertensive therapy. Patients who switched between different β -blockers during the follow-up period were excluded. To ensure sufficient data for diagnosis and healthcare utilization, patients must have had at least two office visits or one hospitalization at the baseline period and at least one healthcare encounter during the one-year follow-up period.

To reduce confounding from health conditions that might cause insomnia, patients with the following diseases at the baseline period were also excluded: chronic obstructive pulmonary disease (COPD), congestive heart failure (CHF), Parkinson's or Alzheimer's disease, anxiety, depression and other psychiatric disorders, chronic pain, urologic diseases [*e.g.*, benign prostatic hyperplasia (BPH) and urinary incontinence (UI)], and hyperthyroidism. In addition, patients who received drugs with CNS stimulant effects were excluded. Furthermore, patients who had any visit related to sleep disorder or received pharmacological treatment with anxiolytics, sedatives, hypnotics, antidepressants, or antipsychotics at the baseline were also excluded.

Definition and classification of exposure: The targeted exposure (the main independent variable) was the use of a specific β -blocker, including propranolol, nadolol, pindolol, metoprolol, betaxolol, carteolol, acebutolol, atenolol, bisoprolol, carvedilol, labetalol, and alprenolol. Exposure duration was estimated from the quantity prescribed, and discontinuation was defined as a lack of refill for more than 30 days after the end date of the previous prescription. The β -blockers were classified in this study according to their selectivity, where alprenolol, pindolol, propranolol, nadolol, carteolol, labetalol, and carvedilol were categorized as the non-selective group, and metoprolol, atenolol, acebutolol, betaxolol and bisoprolol as selective β_1 -blockers.²⁹⁾ The β -blockers were also classified by degree of lipophilicity based on the *n*-octanol/phosphate buffer partition coefficients³⁰⁾ and pharmacokinetic properties,³¹⁾ where atenolol, betaxolol, carteolol, and nadolol were considered to have low lipophilicity, and propranolol, labetalol, carvedilol, bisoprolol, metoprolol, acebutolol, alprenolol, and pindolol to have high lipophilicity.

Study end point: The primary end point was a new insomnia event occurring within 30 days after the index date. The 30-day duration was employed to be consistent with current available information on the insomnia side effect of β -blockers, which is mostly based on previous clinical trials reporting side effects within a short-term timeframe. In this study, an insomnia event was identified by a new diagnosis of sleep disorder or a new prescription of sedatives/hypnotics. Insomnia diagnoses were defined by ICD-9-CM codes of 780.50, 780.52, 780.59, 307.40, 307.41, 307.42, 307.47, 307.48, 307.49, 780.51, 780.57, and sedatives/hypnotics included flurazepam, nitrazepam, flunitrazepam, estazolam, triazolam, temazepam, zopiclone, zolpidem, and zaleplon. To investigate the possible late development of insomnia events, a secondary end point of new insomnia events within 90 days after the index date was also studied.

Statistical analysis: To compare the characteristics between groups, t-tests were performed for continuous variables and chi-square tests were used for dichotomous variables. Logistic regres-



^aAntihypertensive agents include diuretics, β -blockers, CCB, ACEI/ARB, and selected α -blockers and vasodilators acting on smooth muscle.

^bThe baseline period was defined as one year prior to the index date.

^cThe follow-up period was defined as one year from the index date.

Fig. 1. Flow diagram of selecting study cohort

sion was used to analyze the relationship between β -blockers and insomnia. Covariates adjusted in the regression model were baseline characteristics of age, sex, concurrently prescribed antihypertensives (e.g., diuretics, calcium channel blockers, ACEI/ARB, and vasodilators), and comorbid conditions that have been reported as associated with insomnia (e.g., gastroesophageal reflux disease, ischemic stroke, diabetes mellitus, rhinitis, hemodialysis, and ischemic heart disease). Propranolol was used as the reference group to estimate relative risks among β -blockers because it used to be the most commonly prescribed β -blocker in Taiwan.³²⁾ Model fit was investigated with Hosmer and Lemeshow analysis to test the null hypothesis that the data were generated by the fitted model.³³⁾ All statistical analyses were carried out with SAS version 9.2.

Results

A total of 4,063 patients with newly diagnosed hypertension during 2004–2006 who had received β -blocker therapy and met all the inclusion and exclusion criteria were identified (Fig. 1). Compared to propranolol users, patients treated with other β -blockers were younger and more likely to be male. The comorbidities at the baseline were similar between the propranolol users and non-propranolol users except that the non-propranolol users had a significantly lower prevalence of ischemic heart disease. The propranolol users were more likely to receive agents acting on angiotensin and vasodilation (Table 1).

Table 1. Baseline characteristics and medical condition of 4,063 hypertensive elderly patients

β -Blocker therapy	Propranolol user (n = 760)	Non-propranolol user (n = 3,303)	p-value
Demographic variable			
Age, mean (SD), y	73.3 (6.3)	72.3 (6.1)	< 0.001
Age group, y			
65–69	264 (34.7)	1,345 (40.7)	< 0.001
70–74	207 (27.2)	928 (28.1)	
75–79	162 (21.3)	557 (16.9)	
80–84	78 (10.2)	327 (9.6)	
≥85	49 (6.5)	146 (4.4)	
Women (%)	401 (52.8)	1,620 (49.1)	0.06
Medical condition (%)			
GERD	2 (0.3)	8 (0.2)	0.92
Ischemic stroke	31 (4.1)	147 (4.5)	0.65
Diabetes mellitus	84 (11.1)	371 (11.2)	0.88
Rhinitis	16 (2.1)	64 (1.9)	0.76
Hemodialysis	4 (0.5)	15 (0.5)	0.79
Ischemic heart disease	41 (5.4)	101 (3.1)	0.002
Concurrent prescribed antihypertensive agents at index date (%)			
Diuretics	173 (22.8)	890 (27.0)	0.02
CCB	404 (53.2)	1,928 (58.4)	0.01
ACE and/or ARB	321 (42.2)	1,281 (38.8)	0.08
Vasodilator ^a	77 (10.1)	258 (7.8)	0.04
Number of concurrent prescribed antihypertensive agents at index date (%) ^b			
β -Blocker monotherapy	136 (17.9)	587 (17.7)	0.79
Two combined	354 (46.6)	1,485 (45.0)	
Three combined	197 (25.9)	866 (26.2)	
Four combined	65 (8.6)	320 (9.7)	
Five combined	8 (1.1)	45 (1.4)	

GERD, gastroesophageal reflux disease; CCB, calcium channel blocker; ACEI, angiotensin converting enzyme inhibitors; ARB, angiotensin II receptor blocker, i.e., angiotensin II receptor antagonists.

^aVasodilator includes α -blockers and the vasodilator acting on smooth muscle.

^bThe use of β -blocker is not included.

Insomnia events were identified from 214 (5.3%) patients within 30 days and 345 (8.5%) within 90 days after the index date. The logistic regression model showed that non-propranolol users had significantly lower risk of insomnia, after adjusting for the covariates (adjusted OR, 0.47; 95% CI, 0.35–0.63). Using propranolol therapy as the reference, the model found several commonly used β -blockers to have significantly lower risk of insomnia than propranolol. Specifically, bisoprolol had the lowest risk of insomnia (adjusted OR, 0.31; 95% CI, 0.19–0.50), followed by atenolol (adjusted OR, 0.46; 95% CI, 0.33–0.66), while metoprolol and labetalol had a similar risk for insomnia when compared to propranolol. The significant difference in insomnia risk between propranolol users and non-propranolol users remained similar when extending the analytical time-frame to 90-day insomnia events (Table 2). Patients using selective β_1 -blockers showed a significantly lower risk of insomnia than those using non-selective β -blockers (adjusted OR, 0.48; 95% CI, 0.36–0.64). In addition, β -blockers with low lipophilicity agents were found to have lower risk of insomnia than those with high lipophilicity (adjusted OR, 0.72; 95% CI, 0.53–0.96). All logistic regression models met the goodness of fit criterion.

Discussion

We found that the exposure to specific β -blockers was associated with decreased risk of insomnia in this elderly population, and the results were consistent in either a 30-day or 90-day analytical

Table 2. Multivariate association between specific β -blockers and the risk of insomnia among elderly hypertensive patients

Classification	No. event 30-day (%)	30-day insomnia risk, odds ratio [95% CI]		No. event 90-day (%)	90-day insomnia risk, odds ratio [95% CI]	
		Unadjusted	Adjusted [†]		Unadjusted	Adjusted [‡]
Propranolol user	69 (9.1)	Reference	Reference	87 (11.5)	Reference	Reference
Non-propranolol user ^b	145 (4.4)	0.46 [0.34–0.62]***	0.47 [0.35–0.63] ***	258 (7.8)	0.66 [0.51–0.85]**	0.66 [0.50–0.85]**
Pharmacology class						
Selectivity ^c						
Non-selective	105 (8.1)	Reference	Reference	143 (11.0)	Reference	Reference
Selective β_1 blocker	109 (3.9)	0.47 [0.35–0.62]***	0.48 [0.36–0.64] ***	202 (7.3)	0.64 [0.51–0.80]***	0.66 [0.53–0.84]***
Lipophilicity ^d						
High lipophilicity	138 (6.2)	Reference	Reference	210 (9.4)	Reference	Reference
Low lipophilicity	76 (4.2)	0.66 [0.50–0.89]**	0.72 [0.53–0.96]*	135 (7.5)	0.78 [0.62–0.97]**	0.86 [0.68–1.08]
Active ingredient						
Propranolol	69 (9.1)	Reference	Reference	87 (11.5)	Reference	Reference
Bisoprolol	24 (3.1)	0.32 [0.20–0.51]***	0.31 [0.19–0.50]***	50 (6.4)	0.53 [0.37–0.76]***	0.50 [0.35–0.72]***
Atenolol	73 (4.2)	0.44 [0.31–0.62]***	0.46 [0.33–0.66]***	125 (7.2)	0.60 [0.45–0.80]***	0.63 [0.47–0.85]**
Carvedilol	21 (6.1)	0.65 [0.39–1.07]	0.64 [0.38–1.08]	35 (10.1)	0.87 [0.57–1.31]	0.82 [0.54–1.25]
Metoprolol	7 (6.9)	0.75 [0.33–1.67]	0.71 [0.31–1.60]	11 (10.9)	0.95 [0.49–1.84]	0.84 [0.43–1.64]
Labetalol	13 (8.3)	0.90 [0.49–1.68]	0.84 [0.45–1.59]	18 (11.5)	1.00 [0.58–1.72]	0.89 [0.51–1.55]
Others ^e	7 (3.9)	0.40 [0.18–0.89]*	0.41 [0.18–0.91]*	19 (10.4)	0.90 [0.53–1.52]	0.95 [0.56–1.61]

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

[†]Model included age, sex, known GERD, ischemic heart disease, stroke, diabetes mellitus, rhinitis, hemodialysis, and antihypertensive agents such as diuretics, CCB, ACEI /ARB, or vasodilators if they were concurrently prescribed at the index date.

^bNon-propranolol users were composed of nadolol, pindolol, metoprolol, betaxolol, carteolol, acebutolol, atenolol, bisoprolol, carvedilol, labetalol, and alprenolol users.

^cPropranolol, nadolol, pindolol, carteolol, alprenolol, carvedilol, labetalol, and sotalol were categorized as non-selective, and metoprolol, betaxolol, acebutolol, atenolol and bisoprolol were categorized as selective β_1 blockers.

^dAtenolol, betaxolol, carteolol, nadolol, and sotalol were classified as having low lipophilicity, and propranolol, labetalol, carvedilol, bisoprolol, metoprolol, acebutolol, alprenolol, and pindolol were classified as having high lipophilicity.

^eIncluding nadolol, pindolol, carteolol, betaxolol, acebutolol and alprenolol users that had a sample size < 100 .

timeframe. Although sleep disorders associated with β -blockers are considered to be non-urgent and subjective symptoms,^{2–5)} these side effects reduce the patient's quality of life³⁴⁾ and may lead to long-term hypnotic treatments that could in turn trigger an adverse cascade in elderly patients.^{22,35)} To our knowledge, this study is the first to compare the risk of insomnia among elderly patients who were receiving β -blockers for hypertension treatment. Previous clinical trials only reported insomnia side effect of propranolol, pindolol, metoprolol, and atenolol.⁶⁾ This study, using real-world data, provided a more comprehensive analysis among all β -blockers and demonstrated a significant association between specific β -blocker treatment and the incidence of insomnia.

Compared to propranolol users, bisoprolol had the lowest risk of insomnia, followed by atenolol, while the α_1 - and β -blockers such as labetalol and carvedilol had a similar risk for insomnia in our study. The results were consistent with previous studies that compared the insomnia risk of β -blockers by indirect comparisons of clinical trial data and found propranolol to have the highest risk, followed by metoprolol and then atenolol.⁶⁾ Although inconclusive and incomplete, the literature suggests that β -blockers increase the risk of insomnia through centrally mediated mechanisms.³⁶⁾ The centrally mediated mechanism is involved when sufficient amount of β -blockers have penetrated into the brain and bound to β -adrenergic receptors and changed the activity in several networks controlled by β -adrenergics. A central β -adrenergic blocking effect may reduce the signs of rapid eye movement (REM) sleep. β -Blockers could also reduce the Non-REM sleep through suppressing serotonergic pathway. Because the "memory-erasing" Non-REM sleep is eliminated, the probability of remembering dreams would be increased and lead to premature awakening.³⁷⁾ The risk also appears to be determined to some degree by the lipophilicity

and selectivity of β -blockers. McAinsh and Cruickshank had reported in 1990 that the rate of CNS side effects was lower in the less lipophilic β -blockers (*e.g.*, atenolol), as compared to those with higher lipophilicity, such as propranolol, pindolol, and metoprolol.⁶⁾ The *n*-octanol/phosphate buffer partition coefficient has been used to measure the lipophilicity of a compound.^{30,38)} We have performed a simple correlation analysis between the degree of lipophilicity and insomnia risk. The results showed that each additional one unit increase of *n*-octanol/phosphate buffer partition coefficient was associated with a 4% increase in insomnia risk (data not shown). There has also been some discussion on the differential risks of CNS side effects among β -blockers with different selectivity, such as selective (predominant β_1 -blockade) or non-selective (similar β_1 - and β_2 -blockade). Although published data are very limited, studies generally indicated that CNS-related symptoms, including insomnia, associated with non-selective β -blockers were more pronounced, quantitatively or qualitatively, than those associated with β_1 -selective blockers.^{2,15,39)} The current study found that atenolol and bisoprolol had a significantly lower risk than propranolol in both 30-day and 90-day new insomnia events, and that selective β_1 -blockers had a significantly lower risk of insomnia than non-selective β -blockers. It also found that, as a group, β -blockers with low lipophilicity were less likely to have the risk of insomnia (adjusted OR, 0.72), which was consistent with previous literature on the relationship between CNS side effects and lipophilicity. However, this study also found a significant within-group variation in risk of insomnia among β -blockers that have similar lipophilicity. For example, the newer β -blockers with high lipophilicity and higher selectivity, such as bisoprolol, did not show a higher risk of insomnia than the older β -blockers.

Our study suggests that patients using certain β -blockers (*e.g.*,

labetalol and propranolol) as the first-line treatment for controlling blood pressure may suffer a higher risk of insomnia. Although there could be a dose response relationship between the dose of β -blockers and insomnia risk, we were unable to examine this relationship because there were very few dose variations among this study population. Future studies should examine the dose response relationship of insomnia risk of β -blockers. In addition, we did not examine the insomnia risk of other antihypertensive agents. It is possible that other co-medications could also affect the insomnia risk. Considering the fact that elderly patients are already more vulnerable to insomnia and that an adverse cascade could be triggered if hypnotics are prescribed to control insomnia, it is very important for clinical practitioners to be aware of the difference between β -blockers in insomnia risk. Therefore, when a β -blocker is deemed necessary for an elderly patient with hypertension, atenolol or bisoprolol might be a better choice than propranolol or labetalol.

This study has some limitations. There could be certain unmeasured confounding variables in an observational study that might impact the investigation on causality. Although socioeconomic variables, family relationship, and behavioral and environmental factors relevant to insomnia were not available, we did identify the physical illnesses known to cause insomnia and accounted for illnesses in multiple regression models. Confounding by indication where β -blockers with lower risk of insomnia were prescribed to patients who were prone to this side effect, although possible, could be limited in the current study because, according to the literature,⁴⁰⁾ side effects (including the CNS risk) are rarely a consideration to physicians when choosing an antihypertensive agent. When the selection of an exposure is unrelated to the outcome of interest, the bias from confounding by indication is generally minimal.⁴¹⁾ It is also possible that insomnia events might be under-reported by patients or under-diagnosed by physicians. However, since the focus of this study was the comparative risk between β -blockers and there have been no evidence that the under-reporting or under-diagnosis might be associated differentially with any specific β -blockers, there should be no significant bias in the study results due to this limitation. Because the use of ACEI/ARB might not necessarily be related to antihypertensive therapy, the patients who used ACEI/ARB without hypertension diagnosis in the baseline period were not excluded. It is possible that the patients used ACEI/ARB for controlling blood pressure without diagnosis, so additional sensitivity analyses have been conducted by excluding ACEI/ARB patients who might be using the medications for renal protection due to existing health conditions. The consistent results could bolster the robustness of our findings (data not shown). Moreover, several strategies had been undertaken to reduce the potential biases in this study; for example, we restricted the study population to those who did not have diseases known to cause insomnia at the baseline and those with newly diagnosed hypertension to control for the possible difference between the comparison groups in disease severity. The significant association between β -blockers and 90-day insomnia risk also indicates the robustness of the study outcome. Lastly, to accurately identify insomnia events attributable to the use of β -blockers, patients with comorbidities that could cause insomnia, either due to the health conditions or the associated treatments, have been excluded from the study. Namely, we only selected elderly patients who were newly diagnosed with hypertension to examine the insomnia risk caused by certain β -blockers. Readers are cautioned that the

exclusion could affect the generalizability of the study results.

In conclusion, the results of this population-based study demonstrated that the use of bisoprolol and atenolol as first-line treatment was associated with the lowest risk of insomnia in elderly patients, as compared to propranolol. Both the selectivity and lipophilicity of β -blockers appeared to influence the risk of insomnia, in that β -blockers with high selectivity in β_1 -receptors and/or low lipophilicity were associated with a lower risk of insomnia. The results of this study suggest that β -blockers with a low risk of insomnia should be used in elderly hypertensive patients and if at all possible, lower doses should be prescribed if β -blockers with high risk of insomnia are to be indicated for this population.

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